

Train, Validation, Test

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Hold Out Test Data

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Absolute Error and MAE

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Error Analysis

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Assign data roles for Validation and model selection. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Validation and model selection. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Training errors P 2 Q 0 R 1; validation errors P 4 Q 6 R 4. Rule: smallest validation MAE, then simpler; P is simpler than R. Choose and state the test action.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

You have 30 leaves with 3 views each. Allocate 18 leaves to train,6 to validation,6 to test. How many images are in each? Where is a learned normalisation mean fitted?

6. Solve the given case

Split 100 independent leaves into 60 train,20 validation,20 test.

Explain roles and preprocessing.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Assign roles: set A fits model parameters; B selects between two models; C is opened once after choices are fixed. Which is train, validation and test?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 24 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Publish the grouped split strategy and counts for the Plant Classifier, including why plant identity stays within one split.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

You have 30 leaves with 3 views each. Allocate 18 leaves to train, 6 to validation, 6 to test. How many images are in each? Where is a learned normalisation mean fitted?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Choose a candidate

Apply the worked selection rule. Explain why B is not chosen.

My reasoning and evidence

13. W2 Assign data roles

Where should parameter fitting, setting selection and final reporting occur?

My reasoning and evidence

14. W3 Record another attempt

A team creates D after reading validation mistakes. What changes in the record?

My reasoning and evidence

15. W4 Apply independently

Training errors P 2 Q 0 R 1; validation errors P 4 Q 6 R 4. Rule: smallest validation MAE, then simpler; P is simpler than R. Choose and state the test action.

My reasoning and evidence

16. W5 Limit a claim

Why is the winning validation score not a guaranteed future score?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Train, Validation, Test

Assign a separate job to training, validation and final test evidence.

Candidates A and B have training MAE1 and 0 cm; validation MAE2 and 3 cm. Selection rule is smallest validation MAE.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Train, Validation, Test

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit on training

Both candidates are built using training data; B fits it more closely.



2. Select on validation

Choose A because 2 cm is below 3 cm.



3. Reserve final test

Freeze A before scoring final cases; do not select again using their labels.

CHANGE ONE THING

A and B both have validation MAE 2 cm; A is simpler under a predeclared tie rule.

Explain the changed result using the matching lesson.

Why does B's zero training MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Assign a separate job to training, validation and final test evidence.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Freeze A before scoring final cases; do not select again using their labels.

1. Reserve final test
2. Fit on training
3. Select on validation

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Reserve final test Freeze A before scoring final cases; do not select again using their labels. Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: A and B both have validation MAE2 cm; A is simpler under a predeclared tie rule. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Select A using the predeclared simpler-model tie-break.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Select A using the predeclared simpler-model tie-break. The validation errors tie, so the stated second a = ry rule decides without consulting final data.

Repair a missing building block

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as 4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Error Analysis (ai-4/18). Goal: Categorize development errors using evidence. Short practice: What evidence would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
